3.1.1 Atomic structure: Sheet 4

Electron configuration and ionisation energy · 17 marks · about 25 minutes

Key information

- Ionisation energy: energy needed to remove one mole of electrons from one mole of gaseous atoms (or ions).
- Successive IE equations always show gaseous ions losing one electron, eg $Na^+(g) \rightarrow Na^{2+}(g) + e^-$
- Fill sub-levels 1s 2s 2p 3s 3p 4s 3d 4p. Cr and Cu are exceptions ($3d^5 4s^1$, $3d^{10} 4s^1$); for transition metal ions, remove 4s electrons first.
- Across a period: IE increases (more protons, same shielding). Down a group: IE decreases (outer electron further away, more shielding).
- Dips: Group 3 (electron in higher-energy p sub-level) and Group 6 (paired p electrons repel).
- A big jump in successive IEs means the electron is now coming from a shell closer to the nucleus.

Section A: Self-quiz

01 Give the full electron configuration of a fluoride ion (F^-).

[1 mark]

02 The Q^{2+} ion has electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$. Identify Q.

[1 mark]

03 Give an equation, including state symbols, to represent the process that occurs when the second ionisation energy of sodium is measured.

[1 mark]

04 Which element has atoms that contain only two unpaired electrons?

[1 mark]

A cobalt
B magnesium
C nickel
D phosphorus

05 Which Period 3 element has these successive ionisation energies?

	1st	2nd	3rd	4th	5th	6th
IE / kJ mol^{-1}	786	1580	3230	4360	16 000	20 000

[1 mark]

A Al
B Si
C P
D S

Section B: Exam questions

06 This question is about the elements in Period 3.

06.1 Give the full electron configuration of the element in Period 3 with the highest first ionisation energy.

[1 mark]

06.2 Table 3 shows some successive ionisation energies for an element in Period 3.

Table 3

Ionisation number	1	2	3	4	5	6	7	8
IE / kJ mol^{-1}	1000	2260	3390	4540	6990	8490	27 100	31 700

Identify the Period 3 element.

Explain your answer.

[3 marks]

07 This question is about atomic structure.

07.1 Write the full electron configuration for each of the following species: Cl^- and Fe^{2+}

[2 marks]

07.2 State which of the elements magnesium and aluminium has the lower first ionisation energy.

Explain your answer.

[3 marks]

08 Explain why the third ionisation energy of beryllium is much higher than the second ionisation energy of beryllium.

[3 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	$1s^2\ 2s^2\ 2p^6$	Source: 7404/1 Jun 2025 Q03.1	1
02	Mn	Source: 7404/1 Jun 2025 Q03.2	1
03	$Na^+(g) \rightarrow Na^{2+}(g) + e^-$	Source: 7404/1 Jun 2023 Q01.2 Ignore state symbol on electron, even if wrong Allow $Na^+(g) + e^- \rightarrow Na^{2+}(g) + 2e^-$ Allow $Na^+(g) - e^- \rightarrow Na^{2+}(g)$	1
04	**C**	Source: 7404/1 Jun 2025 Q16 nickel	1
05	**B**	Source: 7404/1 Jun 2025 Q14 Si	1
06.1	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$	Source: 7404/1 Jun 2023 Q01.1 and Q01.3	1
06.2	M1 sulfur / S M2 large jump after the sixth electron is removed due to the 7th electron being removed / large difference between ionisation energy 6 and 7 M3 electron removed from the (2p) orbital / (second) energy level / (second) shell which is closer to the nucleus / lower in energy / has less shielding	M3 both ideas needed for mark	3
07.1	$Cl^-\ 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ $Fe^{2+}\ 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$	Source: 7404/1 Jun 2017 Q01.1 and Q01.3 If [Ne] or [Ar] used then max 1 if both correct Ignore $4s^0$	2
07.2	M1 Al M2 (Outer) electron in (3)p sub-level / orbital M3 Higher in energy / further from the nucleus so easier to remove	Mg then CE = 0 M2 not just level or shell Both required for M3. Ignore shielding	3
08	M1 Electron is removed from 1(s) (rather than 2(s)) M2 Lower in energy (than 2s) / less / no shielding / closer to the nucleus M3 Stronger attraction between nucleus and outer electron	Source: 7404/1 Jun 2024 Q02.1 Mark independently	3